



Whole genome sequencing of non-aureus *Staphylococcus* species from poultry barn bioaerosols across Alberta reveals circulation of isolates with multiple antibiotic resistant genes and disinfectant resistant genes

Awais Ghaffar^{a,b}, Karen Liljebjelke^a, Sylvia Checkley^a, Frank van der Meer^a, Muhammad Farooq^a, Heshanthi Herath Mudiyanse^a, Mohamed Faizal Abdul-Careem^{a,*}

^a Faculty of Veterinary Medicine, University of Calgary, Calgary, Alberta, Canada

^b Khan Bahadar Choudhry Mushtaq Ahmed, College of Veterinary and Animal Sciences, Narowal, University of Veterinary and Animal Sciences, Lahore, Pakistan

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ABSTRACT

Objectives: Poultry environment has vast variety of bacterial species and non-aureus *Staphylococcus* species (NASS) are predominantly found. Various NASS are normal inhabitant of gastrointestinal tract and skin of chickens. The study objective was to isolate and identify NASS from bioaerosols of layer chicken barns across Alberta, leading to the phenotypic and genotypic antimicrobial resistance (AMR) profiling of the isolated bacterial colonies.

Methods: A total of 15 barns were sampled in this study with an XMX-CV microbial air sampler. Phenotypic AMR profiles were determined using the Sensititre® broth microdilution method on the standard CMV3AGPF plates and whole genome sequencing (WGS) was conducted to confirm the species and AMR genes.

Results: Seven *Staphylococcus equorum*, 5 *Staphylococcus shini*, 1 *Staphylococcus pseudoxylus*, 1 *Staphylococcus cohnii*, and 1 *Staphylococcus gallinarum* isolates were confirmed and identified by WGS and comparative genomic analysis. One of the *S. equorum* isolate was genetically highly divergent from the remaining isolates of the study. The phenotypic susceptibility profile showed 73.3% (11/15) of the isolates were resistant to lincomycin, 66.7% (10/15) were resistant to tetracycline, 33.3% (5/15) were resistant to streptomycin, 20% (3/15) were resistant to erythromycin, and 6.3% were resistant to gentamicin. A total of 10 antimicrobial resistance genes (ARGs) for antibiotics and 1 gene encoding resistance to disinfectants were detected among the study isolates by WGS.

Conclusions: Normal bioaerosol microflora having ARGs may lead to reduced therapeutic effectiveness of antibiotics in poultry and these ARGs can be a source of serious public health concern.

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1. Introduction

Monitoring of bacteria carrying antibiotic resistant genes from poultry barn bioaerosols can provide an insight into the exposure levels of birds, workers, and surrounding environments to these bacterial pathogens [1]. The poultry environment has a vast vari-

ety of bacterial species and staphylococci are the predominantly found group [2]. In general, Staphylococci are normal inhabitants of skin and mucous membranes of birds and livestock. However, there might be coagulase-negative and coagulase-positive *Staphylococcus* species, both can be a cause of serious infection and have their own clinical significance [3].

Bacteria can shift from litter, soil, water and poultry into the aerosols and non-aureus *Staphylococcus* species (NASS) has been isolated much more than the coagulase-positive staphylococci from poultry house bioaerosols [4,5]. *S. equorum*, and *S. cohnii* have been isolated from air samples in poultry houses in Texas and *S. galli-*

* Corresponding author. Mailing address: Faculty of Veterinary Medicine, University of Calgary, 3330 Hospital Drive NW, Calgary, AB T2N 4N1, Canada.

E-mail addresses: faizal.abdulcareem@ucalgary.ca, mfabdulc@ucalgary.ca (M.F. Abdul-Careem).

narum was found in hen houses bioaerosols in China [6,7]. *S. pseudoxylus* was identified by 16S rDNA sequencing from egg yolk [8]. *S. shinii* has recently been identified as a novel species and it is closely related to *S. pseudoxylus* [9]. *S. xylosus* is among the normal inhabitant coagulase-negative staphylococcal species found predominantly in bioaerosol of poultry. *S. equorum* has been isolated from clinical specimens of humans as well [10,11]. *S. cohnii* can cause urinary tract infections, bacteremia, and sepsis [12]. *Staphylococcus* species are extensively studied and used as model organisms for antimicrobial resistance (AMR) studies [13]. Research focusing on the isolation and quantification of NASS in bioaerosols in poultry houses with ARGs have been reported elsewhere; however, no studies in Alberta have been reported [1,6,7,14]. The data regarding the AMR profiling of gram-positive bacteria across layers barns is scarce in Alberta, although there are various reports against gram negative pathogens [15].

A vast variety of bacteria and antimicrobial resistance genes (ARGs) are present and reported in poultry environment [16]. AMR is not only impacting the treatment and effective prevention of poultry illnesses, but also increasingly a threat to the control of bacterial diseases important in public health [17]. Use of antimicrobials is the major driver for the development of AMR [18]. Antimicrobial use in livestock production creates selective pressure on bacteria, favoring survival of bacteria that carry antimicrobial resistance genes and can survive in the antimicrobial rich environment, ultimately resulting in the occurrence of resistant pathogens. Resistant pathogens can be emitted via ventilation to the surrounding environment, and is a serious one health concern [19,20]. Inhalation can be a potent transfer mechanism of environmentally spread ARGs into various hosts [21,22].

Bacteria possess and acquire resistance by various mechanisms including modification of antibiotic targets, enzymatic inactivation of antibiotic compounds, changes in outer membrane permeability and expression of efflux pumps [23]. Antibiotics residues and bacteria carrying ARGs might be excreted in feces into the litter and housing environment which may ultimately be transferred to humans [24]. Most of these antibiotics are also used in human medicine, so the presence of these genes might have severe public health concerns. This study aimed to determine if NASS can be isolated from bioaerosols from poultry houses across Alberta and then study their phenotypic and genotypic AMR, in order to make recommendations for industry or producers about AMR and bioaerosol surveillance.

2. Materials and methods

2.1. Study area and sampling

This cross-sectional study was conducted in Alberta with convenience sampling from 11 pullet and 4-layer barns between January 2022 and January 2024 (Supplementary Table 1), including repeated sampling from 2 pullet barns. Pullets aged between 7 and 15 weeks and layers aged between 44 and 64 weeks were sampled. The XMX-CV air sampler (Dycor Technologies, Edmonton, AB, Canada) was used to collect the air samples. Sampling was conducted in a total of 15 barns across Alberta. The samples were collected in 5 mL of Tris buffer (VWR®, pH 8.0, 1 Molar Sterile Solution, Radnor, PA, USA). Samples were collected at three different sites along the length of each barn, two near the corners of adjacent walls, and the last one at the top center location of the barn. After every visit, the air sampler was properly cleaned and disinfected, following the manufacturer's recommendations. The collected samples were transported to University of Calgary on ice bags. The barns were located at various distances from 3 to 8 hours' drive from Calgary.

2.1.1. Staphylococcus isolation and identification

Samples were plated on mannitol salt agar for isolation of NASS. One colony from each sample was randomly chosen from the mannitol salt agar plates and further streaked on Tryptic soy agar (Difco™ TSA, BD Biosciences, Franklin Lakes, New Jersey, USA) and biochemical testing was conducted with Staph API strips (bioMérieux, Marcy-l'Étoile, Auvergne-Rhône-Alpes, France). Further confirmation of the isolated bacterial species was conducted by WGS analysis.

2.1.2. Antimicrobial susceptibility testing

Phenotypic AMR profiles of NASS isolates were determined using the Sensititre automated broth microdilution method on the standard CMV3AGPF plates (Thermo Fisher Scientific Inc., Waltham, MA, USA). Antibiotics tested were ciprofloxacin (0.12–4 ug/mL), chloramphenicol (2–32 ug/mL), erythromycin (0.25–8 ug/mL), daptomycin (0.25–16 ug/mL), gentamicin (128–1024 ug/mL), lincomycin (1–8 ug/mL), nitrofurantoin (2–64 ug/mL), linezolid (0.5–8 ug/mL), penicillin (0.25–16 ug/mL), streptomycin (512–2048 ug/mL), tetracycline (1–32 ug/mL), vancomycin (0.25–32 ug/mL), and tigecycline (0.015–0.5 ug/mL). Moreover, Gentamicin (10 ug), Streptomycin (10 ug) and Fosfomycin (200 ug) antibiotic discs were tested against the isolated bacterial colonies. The Clinical and Laboratory Standards Institute (CLSI) breakpoints were used for all available antibiotics while European Committee on Antimicrobial Susceptibility Testing (EUCAST) was used for Nitrofurantoin [25–27].

2.1.3. Genome sequencing, assembly, and average nucleotide identity estimation

Non-*aureus* Staphylococcal species isolates were submitted to the Centre de Diagnostic Vétérinaire, Université de Montréal for DNA extraction and whole genome sequencing on MiSeq illumina (San Diego, California, USA). DNA was extracted by QIAamp® cador® Pathogen Mini Kit by Qiacube apparatus (Qiagen, Hilden, Germany) as per companies' recommendations. Library was prepared by Nextera XT (Illumina, San Diego, CA, USA) for a library size of about 600 bp. The sequencing results were processed and analyzed by Galaxy software [28]. The quality control of Fastq files were accessed with FastQC (Version 0.12.1) [29] and further processed for trimming and adapter removal by fastp (Version 0.23.4) [30]. De novo assembly was done with SPAdes (Version 3.15.5) using the default settings [31]. Genome annotation was conducted by Prokka (Version 1.14.6) with default settings [32]. The species identification was done using kraken2 (Version 2.1.3) in galaxy, KmerFinder 3.2 (<https://cge.food.dtu.dk/services/KmerFinder/>), and type (strain) genome server (<https://tygs.dsmz.de/>). Further investigation to check the genetic relatedness of study isolates with closely related isolates were conducted using Average nucleotide identity (ANI) in galaxy with FastANI (Version 1.3). [33]. The species threshold was set to be >95% similarity with reference strains. A heatmap showing the ANI was then visualized in R [34] with various packages including tidyverse and pheatmap [35,36]. Finds SNP sites (Version 2.5.1) was used in galaxy to find the SNP (Single nucleotide polymorphism) in core genome alignments from Roary output [37], and SNPs were calculated by SNP distance matrix in galaxy (Version 0.8.2).

ARGs from assembled genomes were detected by using ABRIcate (Version 1.0.1) [38], Resfinder [39], Comprehensive Antibiotic Resistance Database [40], National Center for Biotechnology Information (NCBI) [32], and MEGARes [41] with 90% similarity threshold, moreover virulence genes profiling were also screened via virulence factor database [42].

2.1.4. Phylogenetic analysis

The Gff3 files from Prokka outputs of all 15 WGS isolates and 23 closely resembled isolates from NCBI (Supplementary Table 2)

Table 1
Phenotypic AMR profile of 15 NASS.

Antibiotics	Susceptible (%)	Intermediate (%)	Resistant (%)
Tigecycline	100	0	0
Erythromycin	80	6.67	13.3
Tetracycline	33.3	0	66.7
Ciprofloxacin	100	0	0
Chloramphenicol	100	0	0
Penicillin	100	0	0
Daptomycin	100	0	0
Vancomycin	100	0	0
Streptomycin	66.7	0	33.3
Nitrofurantoin	100	0	0
Gentamicin	93.3	0	6.67
Quinpristin/Dalfopristin	100	0	0
Lincomycin	26.7	0	73.3
Linezolid	100	0	0

were processed by Roary (Version 3.13.0) [43] in galaxy with default parameters, to determine core gene alignments. The phylogenetic tree was made with RaxML (Randomised accelerated maximum likelihood) (Version 7.7.6) tree in galaxy with default settings [44]. The phylogenetic tree was then visualized and annotated by using Interactive Tree of Life (iTOL) [30]. The study isolates have been submitted to NCBI under Submission ID: PRJNA1169402.

2.1.5. Ethical considerations

The study experiment protocol was approved by Veterinary Science Animal Care Committee (VSACC) of the University of Calgary, Canada (AC22-0080). Informed consent was obtained from each of the producers before commencement of sampling.

3. Results

In this study, we isolated and confirmed with WGS 1 NASS from each barn included in the study: this included 7 *S. equorum*, 5 *S. shini*, 1 *S. cohnii*, 1 *S. pseudoxylus*, and 1 *S. gallinarum* isolates. ANI analysis proved >97% similarity for nucleotides of all strains of the study with respected reference genomes of the same species and clustering of the species was observed (Fig. 1). Comparative genomic analysis of both *S. shini* and *S. equorum* was conducted, the results showed *S. shini* isolates were at least 99.2% similar based on ANI among each other and 99% similar with one of the reference genomes JBCLTY000000000 (Fig. 2a), suggesting that the isolates were closely resembling among each other. While for *S. equorum* (Fig. 2b), all study isolates were at least 97.2% similar based on ANI among each other, and at least 97.4% similar with UHDI0100001 (supplementary excel sheet, ANI_matrix_sheet). *S. equorum* SAMN44075060 was having at least >33 000 SNPs compared to other study isolates and it was found separate in phylogenetic tree as well suggesting a divergent strains among study isolates (supplementary excel sheet, SNP_distance_matrix_sheet).

Phenotypically, 73.3% (11/15) of isolates were resistant to lincomycin, 66.7% (10/15) were resistant to tetracycline, 33.3% (5/15) were resistant to streptomycin, 20% (3/15) were resistant to erythromycin, and 6.25% were resistant to gentamicin (Table 1). All isolates were susceptible to ciprofloxacin, chloramphenicol, daptomycin, nitrofurantoin, linezolid, penicillin, synercid, vancomycin and tigecycline. Apart from 1, all of the study isolates were phenotypically resistant to at least 1 antibiotic, and 6 isolates were multiple drug resistant (MDR), demonstrating resistance to 3 or more classes of antibiotics.

The genome assemblies of all study isolates are presented in Table 2. The genome size varied from 2 689 132 bp to 3 164 023 bp and contig numbers varied from 44 to 197 (Table 2). Various ARGs detected in each genome are presented in Table 2. The detection rate of *tetK* encoding resistance to tetracyclines was high-

est in 66.7% (10/15) of isolates, followed by *LnuA* encoding resistance to lacosamide's in 66.7% (10/15) of isolates, macrolide phosphotransferase C (*mphC*) gene encoding resistance to macrolides in 66.7% (10/15) of isolates, *str* gene encoding resistance to streptomycin in 33.3% (5/15) of isolates, *msrA* gene encoding resistance to erythromycin in 20% (3/15) of isolates, *fosD* encoding resistance to fosfomycin in 13.1 (2/15) of isolates, *aac* (6')-*aph* (2'') encoding resistance to aminoglycosides in 06.67% (1/15) of isolates, and *fusF* encoding resistance for fusidic acid in 06.7% (1/15) of isolates (Fig. 3). *QacG* conferring resistance to quaternary ammonium compounds was also found in 9 of the study isolates. A total of 10 ARGs for 7 different antibiotics and 1 gene encoding resistance to disinfectants were detected among the study isolates by WGS. The distribution of various ARGs in layers and pullet barns are presented in Fig. 3. Seven isolates carried an *mphC* gene without having phenotypic resistance to erythromycin. Three of the study isolates had phenotypic resistance to erythromycin with both *mphC* and *msrA* genes present in the isolates. No virulence genes were discovered in any of the study isolates.

There were nucleotide substitutions leading to the non-synonymous mutations and changes in amino acid sequences in *mphC* genes as compared to the closely related reference gene (CARD database). The mutations were distributed in *mphC* gene from all 10 isolates at various positions (Fig. 4)

Phylogenetic analysis was conducted for the core genomes of the study isolates and closely resembled isolates. Roary was used for the core genome alignment followed by the maximum likelihood tree by RAXML. Six of the *S. equorum* isolates, including SAMN44075053, SAMN44075054, SAMN44075055, SAMN44075056, SAMN44075057, and SAMN44075061, were closely related to *S. equorum* UHDI01000001, while SAMN44075060 looks divergent from other isolates. All *S. shini* isolates of current study were closely related with JBCLTY000000000 and JAZGMB000000000.1. Isolate SAMN44075047 clustered with *S. pseudoxylus* NZ-CP1626321, SAMN44075058 with *S. cohnii* JAALCY010000001, and isolate SAMN44075059 with *S. gallinarum* QXSC01000001 (Fig. 5).

4. Discussion

The current study was designed to determine if live bacteria, particularly NASS, can be isolated from layer barn air in Alberta, and then, to assess the AMR profiles of NASS phenotypically and genotypically. Consequently, firstly, we discovered *S. pseudoxylus*, *S. shini*, *S. equorum*, *S. cohnii* and *S. gallinarum* circulating in the air of layer barns of Alberta. Secondly, phenotypic and genotypic resistance against currently used antibiotics in humans and animals such as tetracycline, lincomycin, erythromycin, gentamicin, streptomycin, and fosfomycin was observed. Thirdly, we discovered that gene *QacG*, which encodes resistance to disinfectant, was carried in some of the NASS. Finally, some of the characterized NASS showed multidrug resistance.

This is the first report on the isolation of NASS along with the phenotypic and genotypic AMR profiling from layer chicken barn bioaerosols across Alberta, Canada. The cell wall of *Staphylococcus* species has a rigid exoskeleton protecting the bacteria from osmotic lysis and mechanical disruption [45]; however, *Staphylococcus* can survive longer in the bioaerosols. *Staphylococcus* are normal inhabitants of skin and mucous membranes of birds and livestock [3] and have been isolated from skin of animals, (humans), bioaerosols, litter and surfaces [46–48]. The study has isolated and confirmed 15 NASS: 7 *S. equorum*, 1 *S. cohnii*, 5 *S. shini*, 1 *S. pseudoxylus*, and 1 *S. gallinarum*. Various species of NASS have previously been isolated from poultry housing environment [7]. A previous study in Canada has isolated NASS from broiler barn bioaerosols [47]. *S. equorum* has been found to be the mainly iso-

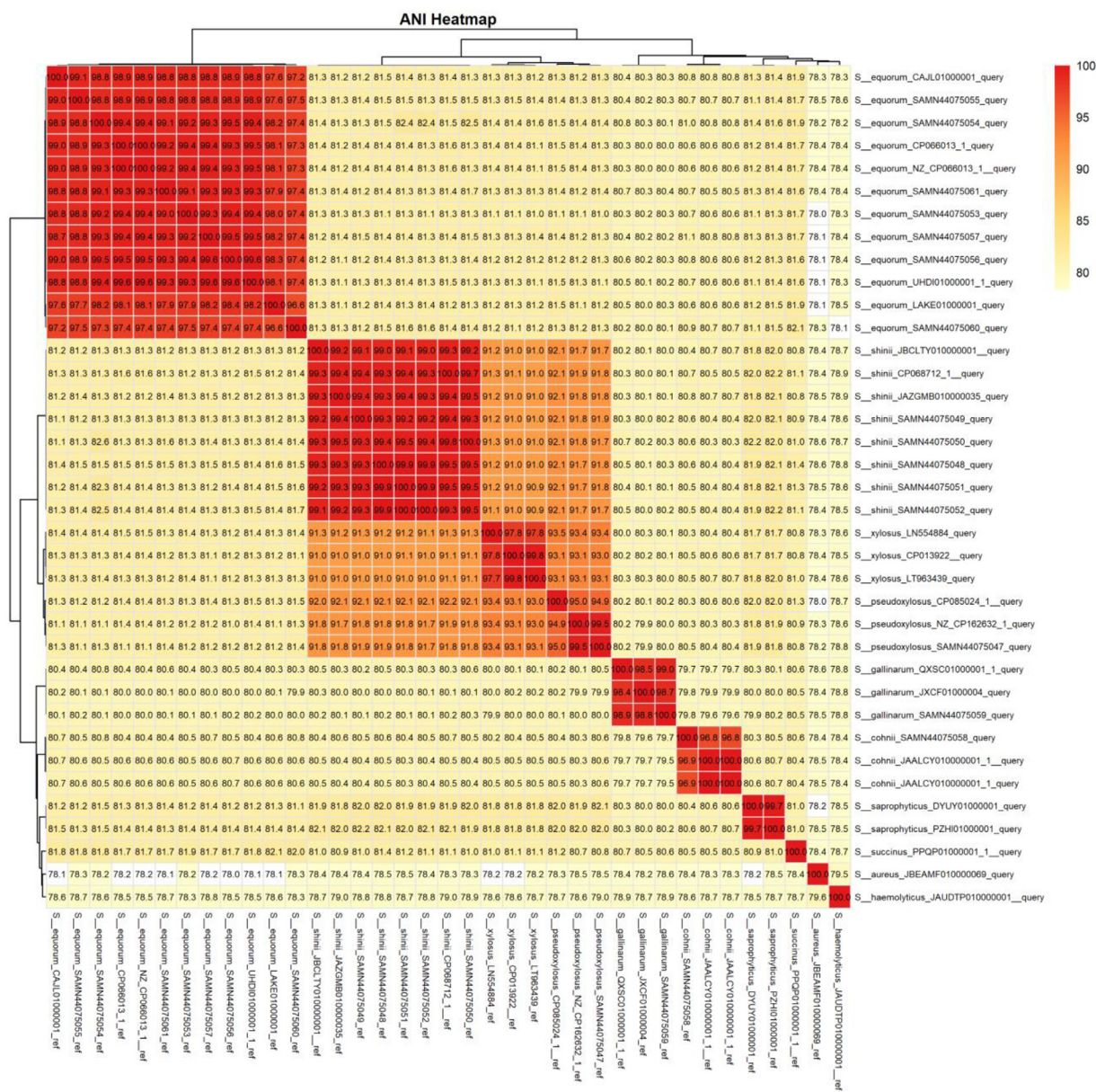


Fig. 1. ANI between study isolates and closely resembled isolates from NCBI.

lated species from egg shells [7]. *S. gallinarum* has been isolated and confirmed from skin of chickens [49]. *S. shinii* has recently been classified as a novel species and some strains of *S. xylosus* and *S. pseudoxylus* have been reclassified to this species [50–52]. *S. cohnii* has been isolated from poultry intestine [53]. Comparative genomics has proved very helpful to better characterize the bacterial species based on WGS data output in this study. The databases for the identification of bacterial species keep on updating with time but care should be taken while using these databases and better perform deeper comparative genomic analysis to identify the bacterial species, particularly with closely related species [51,52].

AMR is associated with the presence of ARGs [54]. Phenotypically, tetracycline and lincomycin resistance were predominantly found in the study isolates. The resistance against these antibiotics was previously reported in Canada [55]. Resistance to tetracycline was explained by the presence of the *tetK* gene which belongs to the major facilitator superfamily of efflux pumps and specifi-

cally encodes for the extrusion of tetracycline from bacterial cells. [56]. *tetK* gene resistance has previously been reported in *S. xylosus* isolates from broiler chicken barns in Canada [57]. Resistance to lincomycin was explained by *lncA* gene which encodes for nucleotidyl transferase which inactivates lincomycin antibiotics [56]. Two of the study isolates also carried the less common *lncG* gene which also encodes for the same enzyme but the genetic sequence is different for both enzymes [58]. The lincomycin resistance with *lncA* gene has previously been reported in *S. xylosus* isolates from broiler chicken barns in Canada. [47].

Macrolide phosphotransferase C (*mphC*) causes inactivation of macrolides antibiotics [59]. The gene was found in ten of the study isolates; 7 of the isolates with *mphC* gene did not have phenotypic resistance to macrolides while the 3 remaining isolates had phenotypic resistance to macrolides but also carried the *msrA* gene. There were various nucleotide substitutions leading to non-synonymous mutations in *mphC* genes in 7 of the study isolates where the gene was present alone without phenotypic expression. These 7 isolates

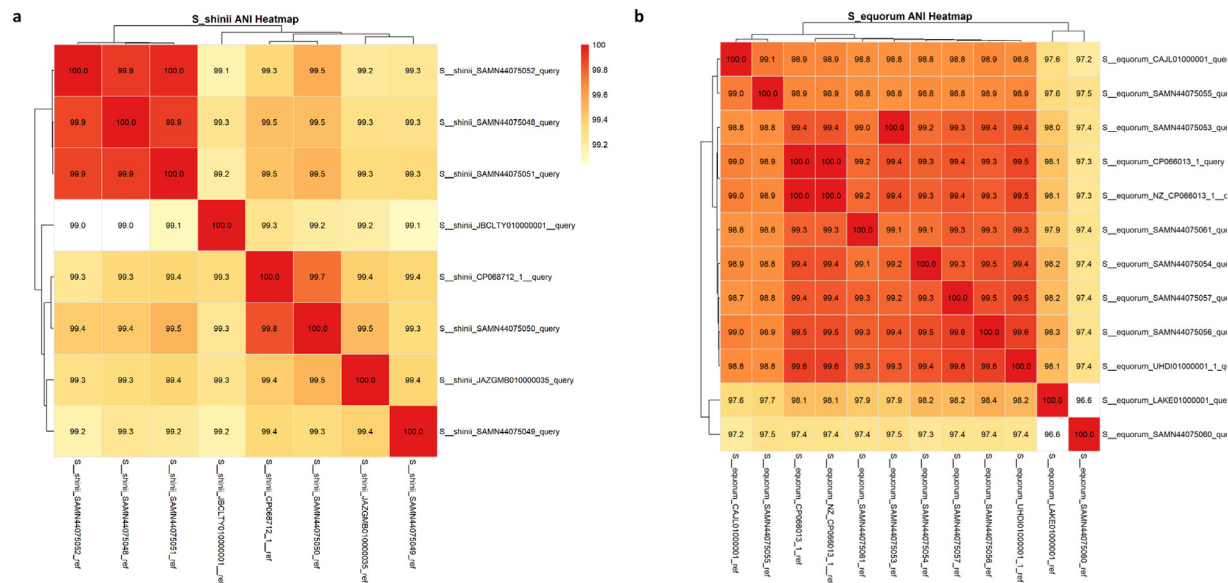


Fig. 2. ANI of *S. shini* and *S. equorum*.

Table 2
Genome characteristics of study isolates along with the ARGs from various databases.

Sample	Genome length (bp)	GC content	No of contigs	Databases for ARGs				Phenotypic resistance
				Resfinder	CARD	NCBI	MEGARes	
<i>S. pseudoxylus</i> SAMN44075047	2 941 742	32.41	48	<i>mphC</i> , <i>LnuA</i> , <i>tetK</i> , AAC (6′)-Ie-APH (2′′)-Ia), ANT4–1b	<i>mphC</i> , <i>LnuA</i> , <i>tetK</i> , AAC (6′)-Ie-APH (2′′)-Ia), ANT4–1b	<i>mphC</i> , <i>LnuA</i> , <i>tetK</i> , AAC (6′)-Ie-APH (2′′)-Ia), ANT4–1b	<i>mphC</i> , <i>LnuA</i> , <i>tetK</i> , ANT4-PRIME, ANT6-PRIME, <i>aadD</i> , QaCG	Lincomycin, Tetracycline, Gentamicin
<i>S. shini</i> SAMN44075048	3 079 027	32.55	126	<i>mphC</i> , <i>LnuA</i> , <i>tetK</i> , <i>FosD</i>	<i>mphC</i> , <i>LnuA</i> , <i>tetK</i> , <i>FosD</i>	<i>mphC</i> , <i>LnuA</i> , <i>tetK</i> , <i>FosD</i>	<i>mphC</i> , <i>LnuA</i> , <i>tetK</i> , QaCG, <i>FosD</i> <i>mphC</i>	Fosfomycin, Lincomycin, Tetracycline
<i>S. shini</i> SAMN44075049	2 985 703	32.52	44	<i>mphC</i>	<i>mphC</i>	<i>mphC</i>	<i>mphC</i>	–
<i>S. shini</i> SAMN44075050	3 066 676	32.46	71	<i>mphC</i> , <i>LnuA</i> , <i>tetK</i>	<i>mphC</i> , <i>LnuA</i> , <i>tetK</i>	<i>mphC</i> , <i>LnuA</i> , <i>tetK</i>	<i>mphC</i> , <i>LnuA</i> , <i>tetK</i> , QaCG	Lincomycin, Tetracycline
<i>S. shini</i> SAMN44075051	3 115 942	32.58	102	<i>mphC</i> , <i>LnuA</i> , <i>tetK</i> , <i>str2</i>	<i>mphC</i> , <i>LnuA</i> , <i>tetK</i>	<i>mphC</i> , <i>LnuA</i> , <i>tetK</i> , <i>str</i>	<i>mphC</i> , <i>LnuA</i> , <i>tetK</i> , ANT6, QaCG	Lincomycin, Tetracycline, Streptomycin
<i>S. shini</i> SAMN44075052	3 164 023	32.48	197	<i>mphC</i> , <i>LnuA</i> , <i>tetK</i> , <i>str2</i>	<i>mphC</i> , <i>LnuA</i> , <i>tetK</i>	<i>mphC</i> , <i>LnuA</i> , <i>tetK</i> , <i>str</i>	<i>mphC</i> , <i>LnuA</i> , <i>tetK</i> , ANT6, QaCG	Lincomycin, Tetracycline, Streptomycin
<i>S. equorum</i> SAMN44075053	2 732 951	32.96	92	<i>lnu(G)_1</i> , <i>str_1</i> , <i>lnu(A)_1</i>	<i>lnu(G)_1</i> , <i>lnu(A)_1</i>	<i>lnu(G)_1</i> , <i>str</i> , <i>lnu(A)_1</i>	<i>QaCG</i> , ANT6, <i>lnu(G)_1</i> , <i>lnu(A)_1</i>	Lincomycin, Streptomycin
<i>S. equorum</i> SAMN44075054	2 735 567	32.92	56	<i>Tetk</i> , <i>lnuG</i>	<i>Tetk</i> , <i>lnuG</i>	<i>Tetk</i> , <i>lnuG</i>	<i>Tetk</i> , <i>lnuG</i>	Tetracycline
<i>S. equorum</i> SAMN44075055	2 779 722	32.87	66	<i>FosD</i> , <i>tetk</i> , <i>lnuA</i>	<i>FosD</i> , <i>tetk</i> , <i>lnuA</i>	<i>FosD</i> , <i>tetk</i> , <i>lnuA</i>	<i>FosD</i> , <i>tetk</i> , <i>lnuA</i>	Lincomycin, Tetracycline, Fosfomycin
<i>S. equorum</i> SAMN44075056	2 689 132	32.98	87	<i>tetK</i> , <i>str2</i>	<i>tetK</i>	<i>tetK</i> , <i>str</i>	<i>Tetk</i> , ANT6	Streptomycin, Tetracycline
<i>S. equorum</i> SAMN44075057	2 735 567	32.92	56	<i>Tetk</i> , <i>msrA</i> , <i>mphC</i>	<i>Tetk</i> , <i>msrA</i> , <i>mphC</i>	<i>Tetk</i> , <i>msrA</i> , <i>mphC</i>	<i>Tetk</i> , <i>msrA</i> , <i>mphC</i>	Erythromycin, Tetracycline
<i>S. cohnii</i> SAMN44075058	2 701 813	32.31	48	<i>msrA</i> , <i>mphC</i> , <i>str2</i> , <i>lnuA</i>	<i>msrA</i> , <i>mphC</i> , <i>lnuA</i> , <i>fusF</i>	<i>msrA</i> , <i>mphC</i> , <i>str</i> , <i>lnuA</i> , <i>fusF</i>	<i>msrA</i> , <i>mphC</i> , ANT6, <i>lnuA</i> , <i>fusF</i> <i>tetK</i> , <i>lnuA</i>	Lincomycin, Streptomycin, Erythromycin
<i>S. gallinarum</i> SAMN44075059	2 994 416	32.99	103	<i>tetK</i> , <i>lnuA</i>	<i>tetK</i> , <i>lnuA</i>	<i>tetK</i> , <i>lnuA</i>	<i>tetK</i> , <i>lnuA</i>	Lincomycin, Tetracycline
<i>S. equorum</i> SAMN44075060	2 818 424	32.94	59	<i>LnuA</i>	<i>LnuA</i>	<i>LnuA</i>	<i>lnuA</i> , QaCG	Lincomycin
<i>S. equorum</i> SAMN44075061	2 772 252	32.84	83	<i>msrA</i> , <i>mphC</i> , <i>tetk</i>	<i>msrA</i> , <i>mphC</i> , <i>tetk</i>	<i>msrA</i> , <i>mphC</i> , <i>tetk</i>	QaCG, <i>msrA</i> , <i>mphC</i> , <i>tetk</i>	Erythromycin, Tetracycline

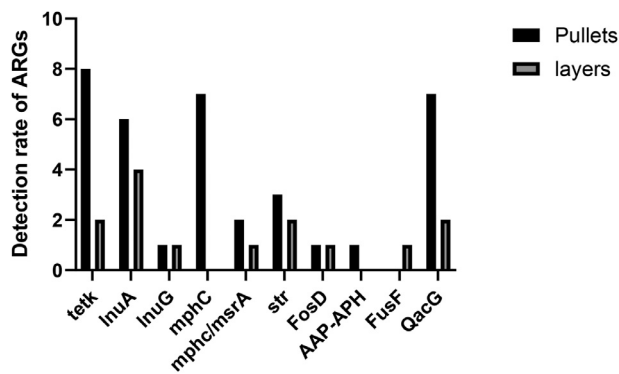


Fig. 3. Distribution of various antimicrobial resistance genes in pullets and layers.

were about 90% similar with the previously reported isolates from CARD database and also the remaining 3 isolates of current study, carrying *mphC* and *msrA* genes, upon alignment among each other. A portion of *msrA* gene is needed for the expression of the *mphC* gene and this gene alone failed to show phenotypic activity against erythromycin [60], potentially explaining the phenotypic and genotypic differences described. This resistance pattern of having ARG without phenotypic resistance, was also observed in susceptible isolates in NASS isolated from dairy herds in Canada and the odds of erythromycin resistance associated with the *msrA* gene was 76.7 (95% CI 12.4–475) and 4.3 for *mphC* (95% CI 2.4–7.8) [54]. The resistance to erythromycin due to *mphC/msrA* has previously been reported in *Staphylococcus* species [61]. Silencing of antibiotic resistance by mutation (SARM) is an important gene silencing mechanism in bacteria [62]. There are previous reports of this mechanism in *S. aureus* where 3.1% of genes were silenced and did not show phenotypic resistance by the SARM mechanism. Mutations in the ARGs silence the expression of respective genes by various mechanism including non-sense mutation, frameshift, loss of translation initiation elements, separation of the promoter region from coding sequence [62,63].

Streptomycin resistance and the *str* gene were detected in 5 of the study isolates. The *Str* gene encodes for aminoglycoside phosphotransferase which causes inactivation of streptomycin and has been reported to causes resistance in *Staphylococcus* species [64]. The resistance to streptomycin has also been previously reported in chicken barn bioaerosols in Canada [47]. One of the study isolates had *fusF* AMR gene in the genome upon sequencing results. The gene encodes for *FusB*-type protein and reported to confer resistance to fusidic acid [65]. The data regarding fusidic acid resistance is scarce in Canada, there are reports of fusidic acid resistance in staphylococci of human origin in Canada [66]. Reports from other countries including South Korea and Portugal suggested the presence of fusidic acid resistance in NASS isolated from chicken samples [66–68]

Two of the study isolates presented resistance to fosfomycin and had *fosD* gene which causes resistance by fosfomycin inactivation [69]. Fosfomycin resistance has been reported in broiler farm and chicken meat from Canada [70,71]. One of the study isolates demonstrated resistance to gentamicin and carried the *aac* (6')-aph (2') gene, which causes inactivation of aminoglycosides [72]. Gentamicin resistance has also been reported from bacteria isolated from chicken samples in Canada [73].

In addition, *QacG* which confers resistance to ethidium bromide and benzalkonium chloride was found in 9 of the study isolates by WGS and the gene has been reported from *S. aureus* isolated from dogs in Canada [74]. There were some genes identified through WGS in this study, where the isolates were either not phenotypically screened or no phenotypic resistance was expressed.

The carriage of ARG by commensal bacteria can be a cause of concern in poultry production as the barns are cleaned and disinfected after every flock, and having those resistance genes in commensal pathogens may be problematic later in the next flock as bacteria usually transmit various ARG by horizontal gene transfer to other bacteria [75]. It will not only have consequences for poultry but is also a major One Health concern. Due to the Canadian Integrated Program for Antimicrobial Resistance Surveillance implementation, there is reduction in AMR of various poultry pathogens [15,76]. This study has found co resistance of various ARGs against antibiotics and commonly used disinfectants. Co-resistance of an-

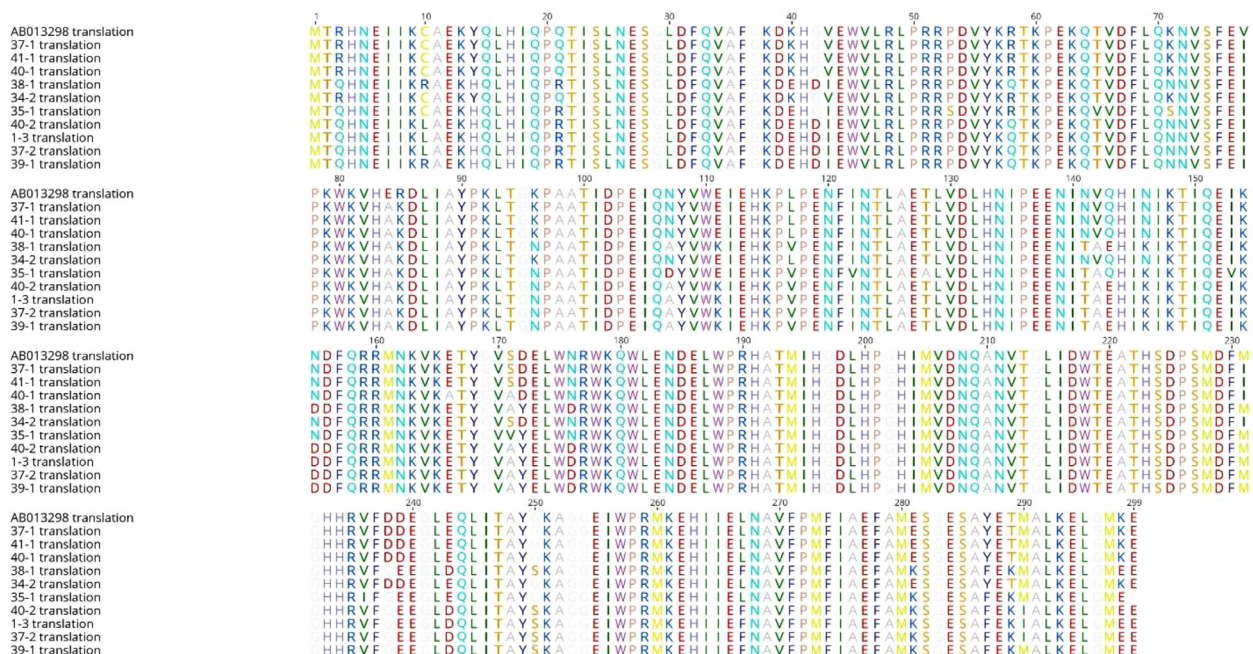


Fig. 4. Amino acid alignment of mphC genes showing non-synonymous mutations among study isolates. Changes in color shows mutations.

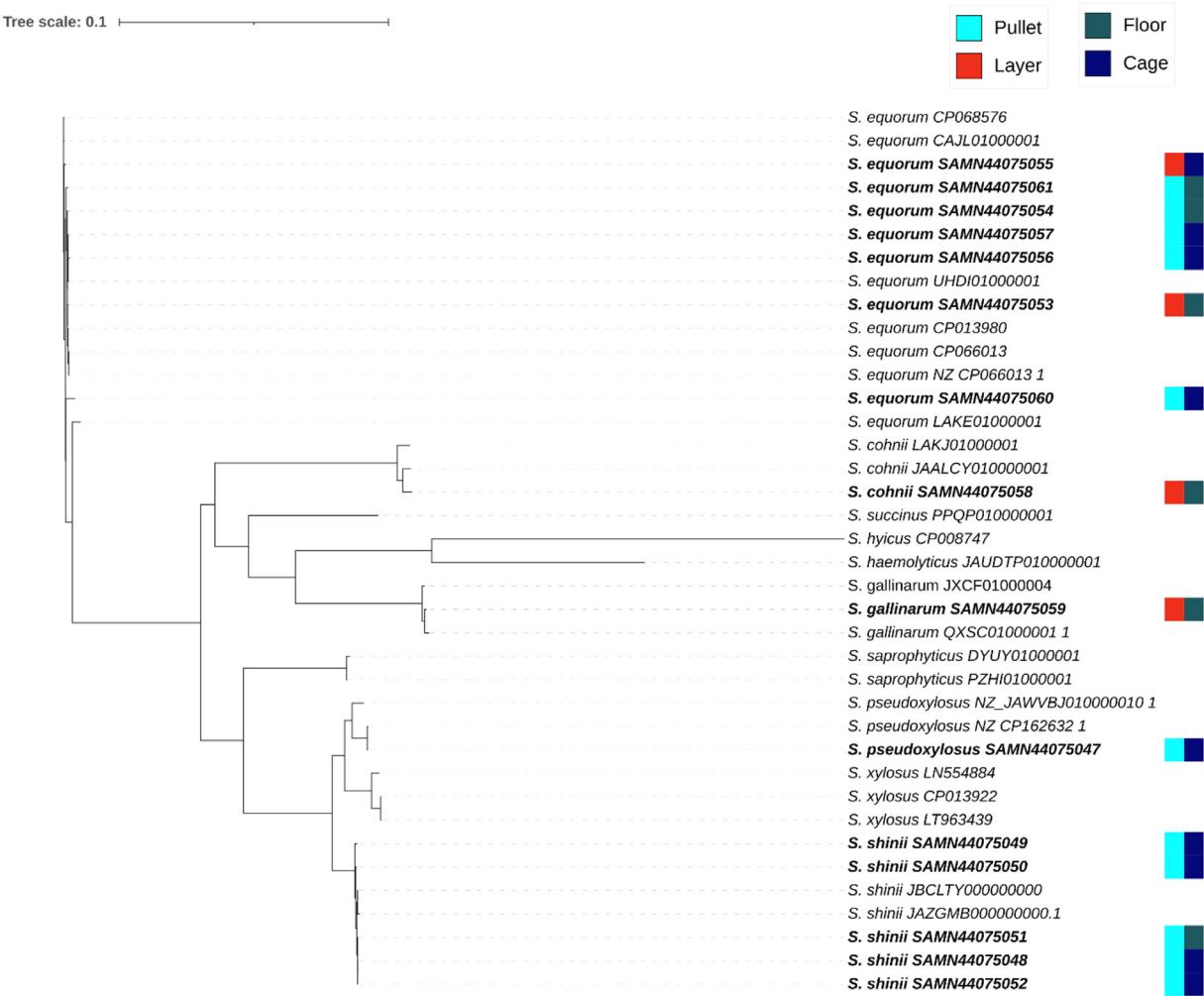


Fig. 5. Core genome based phylogenetic tree of the study isolates (bold ones) and closely resembled isolates from NCBI. The study isolates from the various types of housing systems including cage housed and floor housed and the age of the birds including pullets and layers was attributed by various colours to better see the relationship of the study isolates.

tibiotic and *QacG* has been reported in this study which is previously documented [77]. Six out of 15 isolates had MDR in this study and there are various reports having such MDR isolates from poultry houses in Canada [66,78,79]. In current study according to the background information taken from participant producers (Supplementary Table 1), there was no history of the use of antimicrobials in any of the sampling barns until sampling was conducted,; however, resistance to various antibiotics was observed. The poultry houses where this bioaerosol sampling occurred were all located on large farming operations with multiple other animal species nearby. A lot of pathogens can travel via air to nearby farms [21]. Moreover, wild birds can also transit various ARG to or from poultry [80,81]. Various antibiotics are used for treatment or prevention of infections in poultry, which causes selection pressure and results in the higher prevalence of bacteria with ARGs [82]. Previous studies have reported the association of the use of antimicrobials with antimicrobial resistance in bacteria [83,84]. In Alberta, there are cases of duodenal necrosis in layers which suggests use of antimicrobials [85]. Apart from that, there are other bacterial and viral diseases, in which depending upon the case and scenario, various antibiotics are used [55,85]. Moreover ARGs can be vertically transmitted as microbiota is acquired from the parent flocks and potentially from the environment [86,87]. Having ARGs can not only cause the treatment failure in birds upon using

the same antibiotics for which the ARGs are present, but also increase the disease risk and economic losses to the producers [88]. The identification of ARG against antimicrobials of high importance and medium importance in human health is also a significant One Health concern. Aerosols can play an important role in the transmission of bacteria carrying ARGs between animals and humans, posing a serious emerging one health concern [89].

5. Conclusions

The current study has isolated and confirmed the NASS from layer barn bioaerosols and observed phenotypic and genotypic resistance against currently used antibiotics such as tetracycline, lincomycin, erythromycin, gentamicin, streptomycin, and Fosfomycin. Moreover, we recorded silent antibiotic-resistant gene, *mphC*, and *QacG* that encodes resistance to disinfectant. Further, some NASS isolates showed multidrug resistance traits, and resistance to both antibiotics and disinfectants. WGS based surveillance proved to be a more precise technique for surveillance of AMR and should be used in surveillance studies. Role of bioaerosols is critical as they can be a potent source of AMR transmission. The strains isolated from bioaerosols carry diverse MDR isolates and ARGs; as such the findings have implications not only for poultry production but also for public health.

Animal use approval: The sampling of poultry barn environment was approved by Veterinary Science Animal Care Committee of the University of Calgary (animal protocol # AC22-0080 and the approval date is 7 July 2022).

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Ethical approval: Not required.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.jgar.2025.03.014](https://doi.org/10.1016/j.jgar.2025.03.014).

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